Calorimetry lab

antocciphysics | TEACHER KEY | CPA

Learning target: Use an energy balance to analyze mixing and heat transfer.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 2; 3 course days later.*
$Q=mc\Delta T=10500$ J, assuming the supplied energy heats the sample.

02. Explain a worked example (worked)
Hot loses 10°C worth, cold gains 10°C worth; equal mc makes energy changes equal/opposite.

03. Look and explain (model)
40°C; equal heat capacities, equal temperature changes in opposite directions.

04. Build the solution (practice)
300 J; with cup gain 50 J, cold sample gains 250 J.

05. Use the idea (reason)
Preserve raw measurements; delayed reading allows more exchange with surroundings, biasing the mixing result.

06. Choose a method (mixed) *Review: Unit 4, Day 15; 12 course days later.*
144 J; spring energy scales with compression squared.

07. Exit check (exit)
No; equilibrium should lie between initial temperatures under stated assumptions. Check reading or unmodeled energy source.

08. STRETCH: Apply it another way (stretch)
Above 35°C; weighted mean 40°C.

Teaching note: Lab companion. Use teacher-approved temperatures and handling procedures. Supplied values are illustrative; record actual trials separately.

Calorimetry lab

antocciphysics | TEACHER KEY | HONORS

Learning target: Use an energy balance to analyze mixing and heat transfer.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 2; 3 course days later.*
$c=Q/(m\Delta T)=1000$ J/(kg·°C). No phase change and no unaccounted loss/input.

02. Explain a worked example (worked)
Hot loses 10°C worth, cold gains 10°C worth; equal mc makes energy changes equal/opposite.

03. Represent and interpret (model)
40°C; equal heat capacities, equal temperature changes in opposite directions.

04. Solve and justify (practice)
$(.10\times80+.30\times20)/.40=35$°C.

05. Analyze the reasoning (reason)
Preserve raw measurements; delayed reading allows more exchange with surroundings, biasing the mixing result.

06. Choose a method (mixed) *Review: Unit 4, Day 15; 12 course days later.*
$|x|=\sqrt{(2\ U/k)}=0.06$ m; $|F|=24$ N.

07. Exit check (exit)
Water gains 8400 J; metal $c=8400/(.20\times60)=700$ J/(kg·°C).

08. STRETCH: Apply it another way (stretch)
Water gains 4200 J; metal cools 75°C; $c=4200/(0.1\times75)=560$ J/(kg·°C).

Teaching note: Lab companion. Use teacher-approved temperatures and handling procedures. Supplied values are illustrative; record actual trials separately.